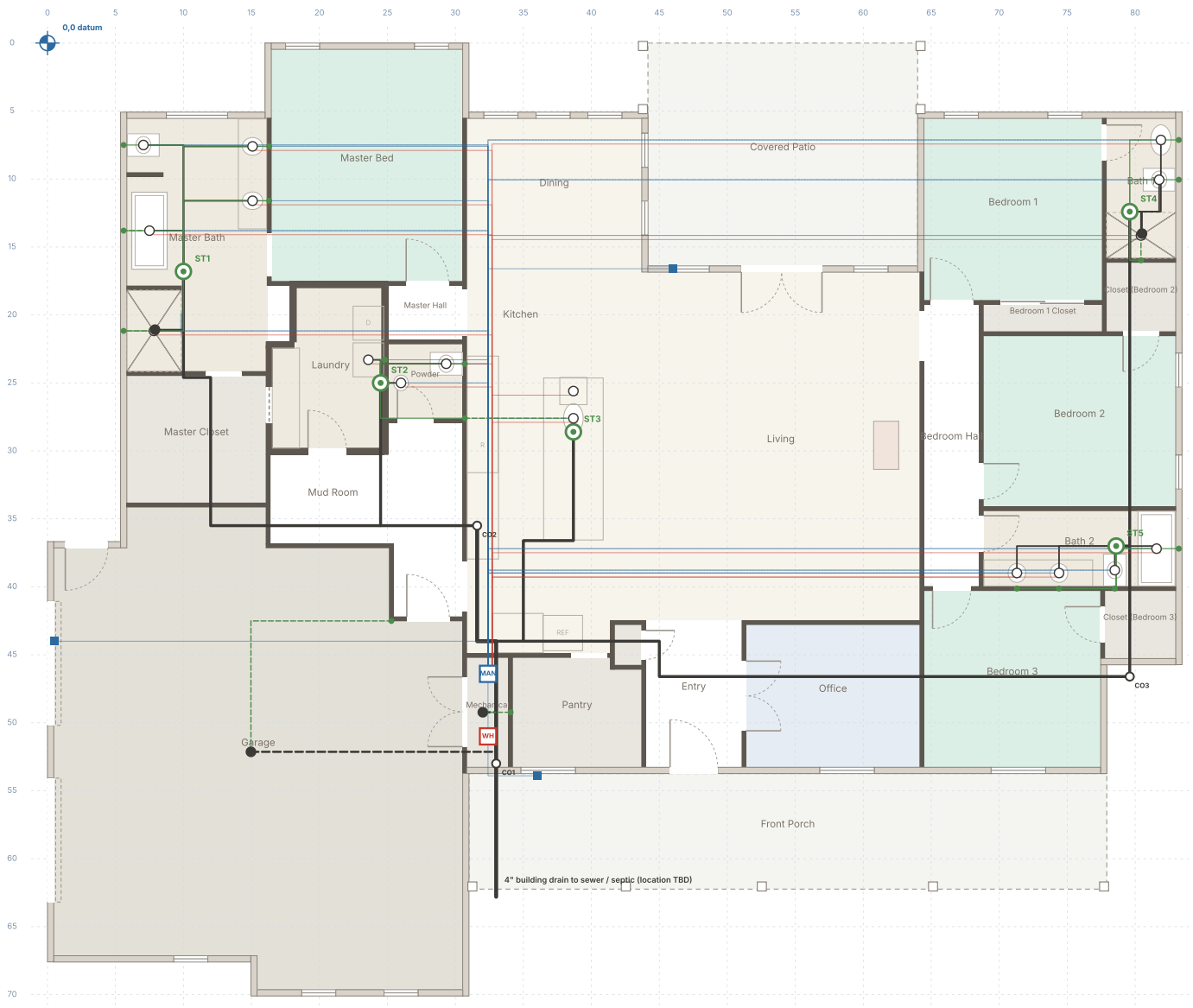
REAR · PATIO



FRONT · PORCH

Drains black (4" main heavy), floor-drain lines dashed, stacks green. · Vents green: a dot where each rises in the wall, thin line across the attic to its stack, dashed where it runs under the slab. · PEX supply home runs: cold blue, hot red. · Routing is schematic: the plumber lays out the final runs.

draft

P-1 Plumbing · 64 Stone Sheep Circle

Fixtures from A-5; drain, vent and supply routing is a first draft for the plumber (slab on grade, 4" building drain out the front). App version v17. From the owner's plans (Larson House, 6/11/26).

Notes

1. Slab on grade assumed (A-5 shows floor drains in the garage and mechanical closet). All drains and supply runs are under the slab or in walls; no attic plumbing.
2. Building drain 4" at 1/4" per ft, 41 DFU total (41 of 216 allowed on 4"). Five fixture groups, each on a 3" stack with a 3" vent through the roof; the kitchen island on a loop vent or an air admittance valve where the AHJ allows.
3. Supply: 1" service to a PEX home-run manifold in the mechanical closet; 25.3 cold + 10.4 hot WSFU. 1/2" PEX to each fixture, 3/4" to hose bibbs. Pressure-balanced or thermostatic valves at the tubs and showers.
4. Wet walls framed 2×6 (see S-1). ADA toilets as drawn: 17-19" bowl height, grab-bar blocking.
5. Water heater in the mechanical closet with a pan and the T&P and pan drain to the closet floor drain.

Open items

- Where the house connects: city sewer or septic, and which side of the lot. The drain is drawn leaving under the front porch.
- Water source (city or well) and where it enters; the manifold is drawn in the mechanical closet.
- Water heater fuel and type. A heat-pump water heater needs about 700 cu ft of air or a ducted kit; the 3' x 8' mechanical closet is too small without one. A gas unit needs venting and combustion air.
- Powder room sink is not drawn on A-5 (one is shown here, assumed).
- Garage floor drain: sanitary with an interceptor, or to daylight (lot grading).
- Kitchen island vent method (loop vent vs. AAV) depends on the local code.
- Fixture models: A-5 tags the drains K-9135 and K-9136; tub, toilet, faucet and valve models still to pick.

Fixture schedule

Tag	Fixture	Room	Trap / drain	DFU	Supply	Stack
MB-WC	ADA toilet	Master Bath	3"	3	1/2"	ST1
MB-TUB	Pedestal tub (faucet)	Master Bath	1-1/2"	2	1/2"	ST1
MB-LAV1	Vanity sink 1	Master Bath	1-1/4"	1	1/2"	ST1
MB-LAV2	Vanity sink 2	Master Bath	1-1/4"	1	1/2"	ST1
MB-SHR	Shower 4' x 6' (glass door)	Master Bath	2"	2	1/2"	ST1
MB-FD	Shower drain K-9136	Master Bath	2"	2	—	ST1
B1-LAV	Vanity sink	Bath 1	1-1/4"	1	1/2"	ST4
B1-WC	ADA toilet	Bath 1	3"	3	1/2"	ST4
B1-SHR	Shower 5' x 3'-4"	Bath 1	2"	2	1/2"	ST4
B1-FD	Shower drain K-9135	Bath 1	2"	2	—	ST4
PR-WC	ADA toilet	Powder	3"	3	1/2"	ST2

Tag	Fixture	Room	Trap / drain	DFU	Supply	Stack
PR-LAV	Lavatory (not drawn on A-5; assumed)	Powder	1-1/4"	1	1/2"	ST2
LA-WASH	Washer (curved front loading)	Laundry	2" standpipe	3	1/2" box	ST2
K-SINK	Kitchen sink (island)	Kitchen	1-1/2"	2	1/2"	ST3
K-DW	Dishwasher (island)	Kitchen	via sink tailpiece	2	3/8" hot	ST3
B2-LAV1	Vanity sink 1	Bath 2	1-1/4"	1	1/2"	ST5
B2-LAV2	Vanity sink 2	Bath 2	1-1/4"	1	1/2"	ST5
B2-WC	ADA toilet	Bath 2	3"	3	1/2"	ST5
B2-TUB	Standard tub 1 [66W]	Bath 2	1-1/2"	2	1/2"	ST5
ME-FD	Floor drain K-9136	Mechanical	2"	2	—	FD line
GA-FD	Floor drain K-9136	Garage	2"	2	—	FD line
HB1	Hose bibb, front porch (frost-free)	Outside	—	—	3/4"	—
HB2	Hose bibb, rear patio (frost-free)	Outside	—	—	3/4"	—
HB3	Hose bibb, garage door wall (frost-free)	Outside	—	—	3/4"	—

Stacks and vents

Stack	Group	Fixtures	DFU	Vent
ST1	Master bath	6	11	3" VTR
ST2	Laundry and powder	3	7	3" VTR
ST3	Kitchen island (island vent)	2	4	loop vent / AAV
ST4	Bath 1	4	8	3" VTR
ST5	Bath 2	4	7	3" VTR

Bill of materials

Rough-order quantities from the routing above, for pricing. The plumber takes off the final list.

Item	Qty	Notes
4" PVC DWV	33 ft	building drain to the front wall + 10 ft to the property line stub (more to the sewer/septic)
3" PVC DWV	206 ft	branches and toilet arms, + 15 %; plus ~50 ft for the four 3" vent stacks
2" PVC DWV	86 ft	showers, washer, floor drains, + vents
1-1/2" PVC DWV	108 ft	lavs, tubs, kitchen sink, + vents
1/2" PEX	1804 ft	home runs cold + hot, + 20 %
3/4" PEX	107 ft	hose bibbs
PEX manifold	1	19 cold + 13 hot ports

Item	Qty	Notes
Water heater	1	50 gal, pan, expansion tank
Toilets	4	ADA height (A-5)
Tub/shower valves	4	pressure-balanced
Shower drains	2	K-9136 (master), K-9135 (bath 1) per A-5
Floor drains	2	garage and mechanical, K-9136 per A-5
Cleanouts	3	4" at the exit, 3" at changes of direction
Hose bibbs	3	frost-free